Vijay Academy Presents

Savitribai Phule Pune University (SPPU)

Data Science and Big Data Analytics (DSBDA)

TE Computer Engineering (2019 Pattern)

One Shot Lecture

Unit 6. Data Visualization and Hadoop

What You Get : Strategy For Exam :

- Simple Explanation
- PPT + Notes
- PYQs + Most IMP Questions

Unit 6. Data Visualization and Hadoop

Syllabus Topics :

1. Introduction to Data Visualization
2. Challenges to Big Data Visualization
3. Types of Data Visualization
4. Data Visualization Techniques
5. Visualizing Big Data
6. Tools used in Data Visualization (Tableau, Power BI etc.)
7. Hadoop Ecosystem
8. MapReduce
9. Pig
10. Hive
11. Analytical Techniques used in Big Data Visualization
12. Line Plot (Python)
13. Scatter Plot (Python)
14. Histogram (Python)
15. Density Plot (Python)
16. Box Plot (Python)

1: Introduction to Data Visualization

What is Data Visualization?

Data Visualization is the process of **representing data in a graphical or visual format** such as charts, graphs, maps and plots so that humans can easily understand patterns, trends and insights.

Simply put — converting raw numbers and data into pictures that are easy to understand.

Why is Data Visualization Important?

Easy to find **patterns and trends** in large data

Helps in **faster decision making**

Makes complex data **easy to communicate**

Helps detect **outliers and anomalies**

Objectives of Data Visualization

Explore and understand data

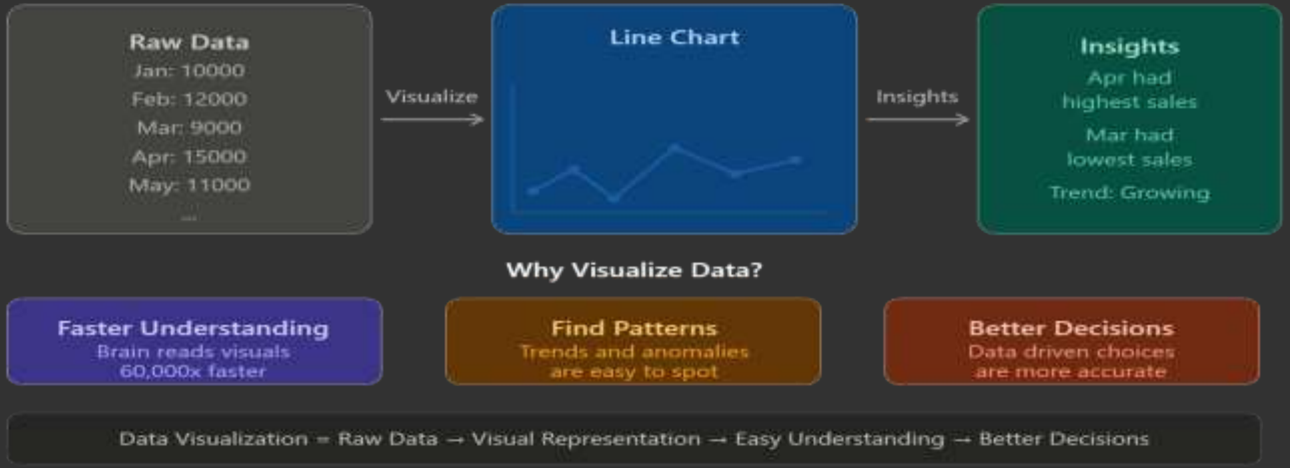
Find hidden patterns and relationships

Support data driven decision making

Monitor performance over time

Characteristics of Good Data Visualization

Characteristic	Meaning
Clear	Easy to understand at a glance
Accurate	Represents data correctly
Simple	Not overcrowded with information
Relevant	Shows only useful information
Interactive	User can explore the data

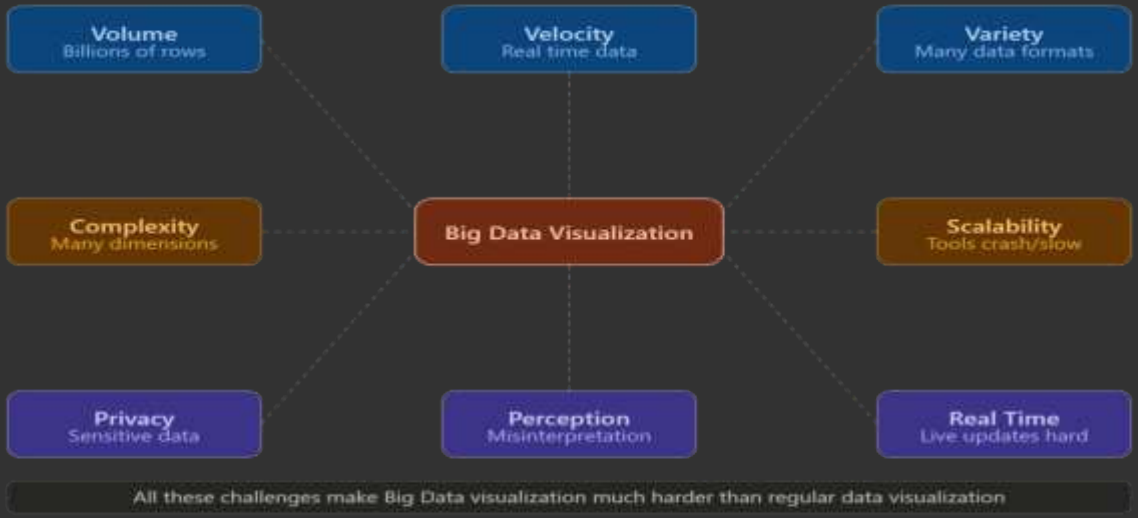


2: Challenges to Big Data Visualization

What are the Challenges?

When data is extremely large (Big Data), visualizing it becomes very difficult. There are many problems that arise when trying to create meaningful visuals from Big Data.

Simply put — normal visualization tools and techniques fail when data is too large, too fast or too complex.



1. Volume
Big Data has **extremely large amounts of data** — billions of rows.
Normal charts cannot display billions of data points
Screen has limited pixels — cannot show every data point
Example: Showing all tweets in a day (500 million tweets) on one graph

2. Velocity
Big Data arrives **very fast** in real time.
Visualization must update in real time
Old tools are too slow to keep up
Example: Stock prices changing every millisecond

3. Variety
Big Data comes in **many different formats** — text, images, videos, numbers, JSON etc.
Hard to visualize different types together
Need different techniques for each type
Example: Combining sales numbers with customer reviews
and social media posts

4. Complexity of Data
Big Data has **many variables and dimensions**.
Hard to show 100 variables in one chart
Relationships between variables are complex
Example: Patient health data with 200 different measurements

5. Scalability
Visualization tools must **scale up** to handle growing data.
As data grows, tools become slow or crash
Hard to maintain performance with huge datasets

6. Privacy and Security
Big Data often contains **sensitive personal information**.
Cannot show private data in visualizations
Example: Patient names and medical records

7. Perception and Interpretation
Even if visualization is created, **humans may misinterpret** it.
Too much information on one chart = confusion
Misleading scales or colors can give wrong impression
Example: A graph with wrong Y-axis scale making small changes look huge

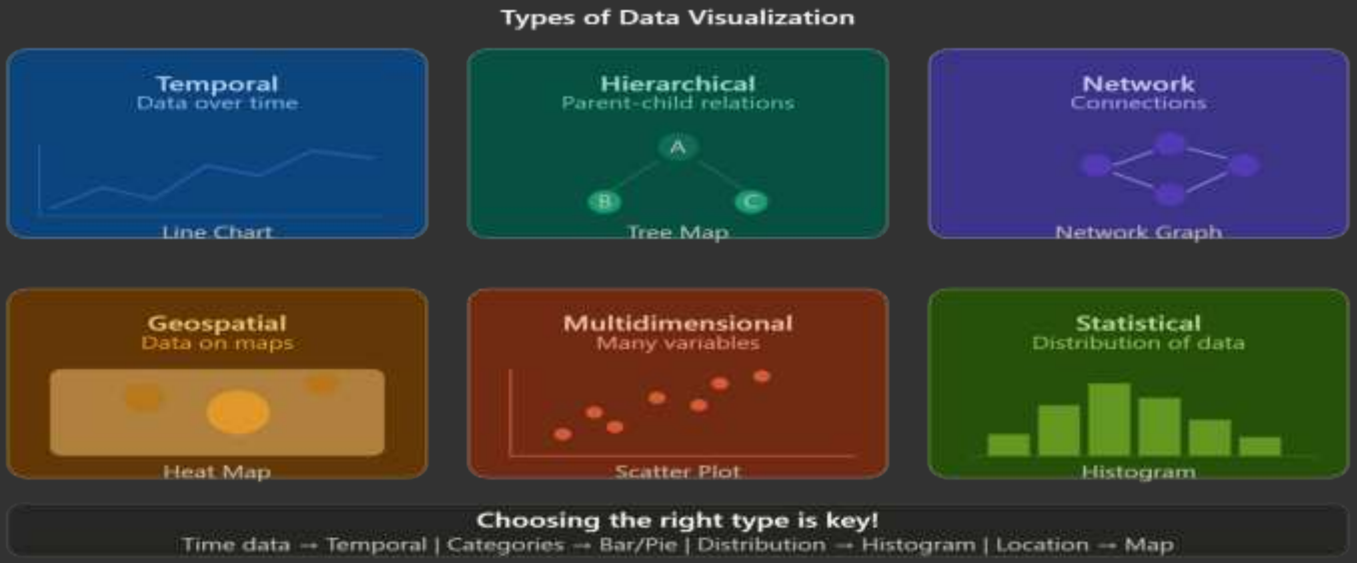
8. Real Time Visualization
Showing **live updating data** is technically very hard.
Requires fast processing and rendering
Tools must handle continuous data streams
Example: Live traffic monitoring dashboard

3: Types of Data Visualization

What are Types of Data Visualization?

Different types of data require different types of visualizations. Choosing the **right type** of visualization is very important to communicate data clearly.

Simply put — just like you use different tools for different jobs (hammer for nails, screwdriver for screws), you use different chart types for different data.



3: Types of Data Visualization

1. Temporal Visualization

Used for data that **changes over time**.
Shows trends over time periods
X axis = time, Y axis = value
Examples: Line chart, Area chart, Timeline
Use case: Stock prices over 1 year, monthly sales data

2. Hierarchical Visualization

Used to show **parent-child relationships** or nested structures.
Shows how things are organized in levels
Examples: Tree map, Sunburst chart, Org chart
Use case: Company org chart, file system structure

3. Network Visualization

Used to show **relationships and connections** between entities.
Shows how things are connected to each other
Examples: Network graph, Node-link diagram
Use case: Social network connections, web page links

3: Types of Data Visualization

4. Geospatial Visualization
Used to show **data on maps** based on location.
Shows geographic patterns and distributions
Examples: Choropleth map, Heat map on map, Bubble map
Use case: Population density by state, COVID cases by district

5. Multidimensional Visualization
Used to show **data with many variables** at once.
Shows relationships between multiple dimensions
Examples: Scatter plot, Histogram, Box plot, Pie chart, Bar chart
Use case: Comparing sales across products, regions and time together

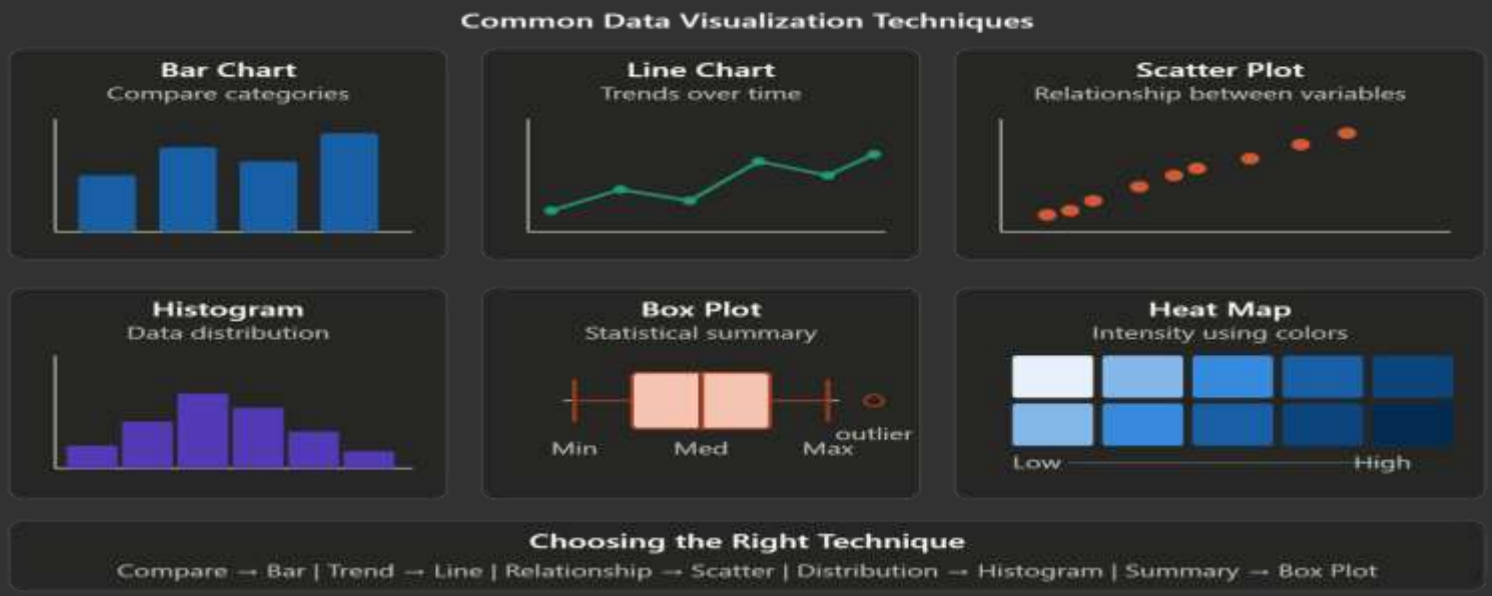
6. Statistical Visualization
Used to show **statistical properties** of data.
Shows distribution, spread and summary of data
Examples: Histogram, Box plot, Density plot, Violin plot
Use case: Showing distribution of student marks

4: Data Visualization Techniques

What are Data Visualization Techniques?

Data Visualization Techniques are **specific methods and approaches** used to represent data visually in a clear and meaningful way.

Simply put — these are the actual "how to draw" methods used to create effective visualizations.



Major Data Visualization Techniques :

1. Bar Chart

Used to **compare quantities** across different categories.
Horizontal or vertical bars
Length of bar = value
Easy to compare multiple categories
Use case: Comparing sales of different products

2. Line Chart

Used to show **trends over time**.
Points connected by lines
Shows increase, decrease or fluctuation
Use case: Stock prices over 6 months, temperature over a week

3. Scatter Plot

Used to show **relationship between two variables**.
Each point = one data record
X axis = variable 1, Y axis = variable 2
Shows correlation (positive, negative, none)
Use case: Relationship between study hours and marks

4. Histogram
Used to show **distribution of a single variable**.
Similar to bar chart but for continuous data
X axis = data range (bins), Y axis = frequency
Shows how data is distributed
Use case: Distribution of student marks in a class

5. Box Plot
Used to show **statistical summary** of data.
Shows Minimum, Q1, Median, Q3 and Maximum.
Shows outliers clearly
Use case: Comparing salary distribution across departments.

6. Heat Map
Used to show **intensity of values** using colors.
Color intensity = value (dark = high, light = low)
Good for showing patterns in 2D data
Use case: Website click patterns, weather temperature maps

5: Visualizing Big Data

What is Visualizing Big Data?

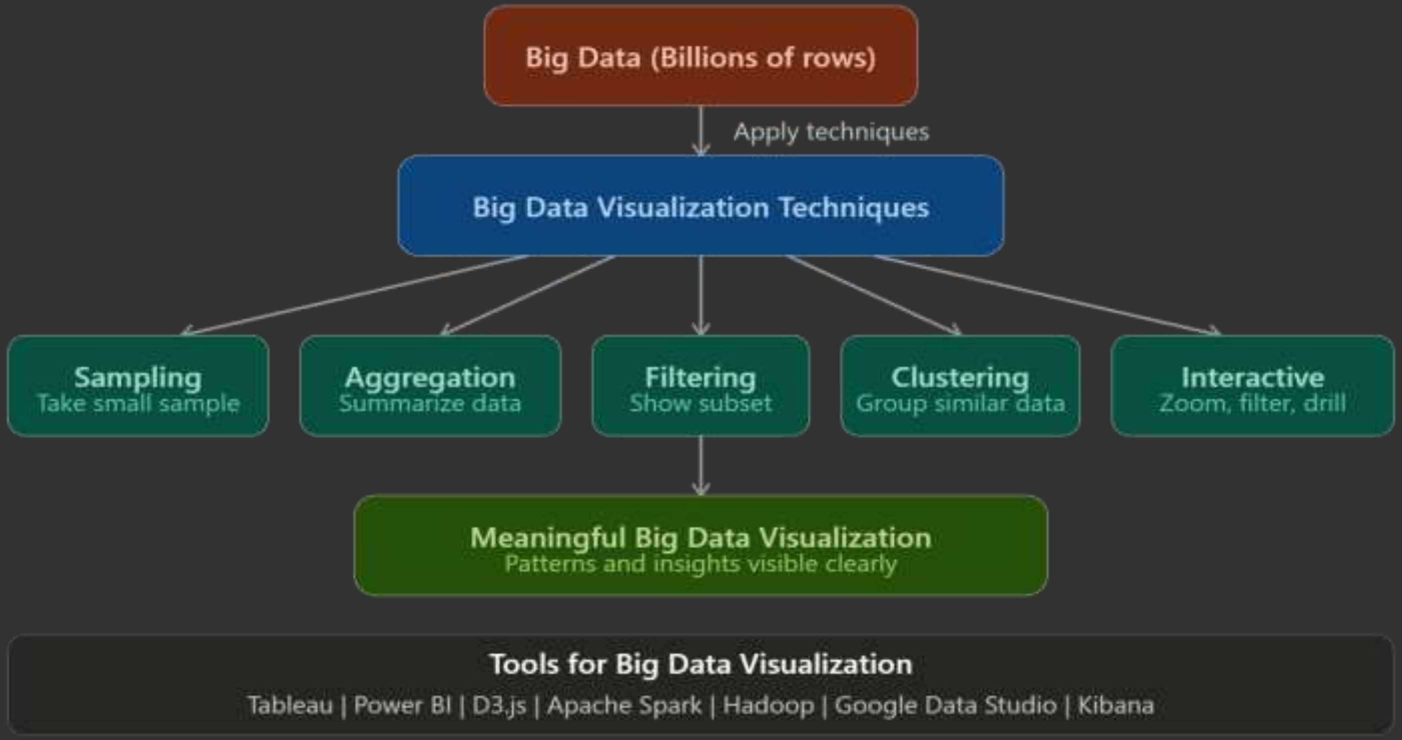
Visualizing Big Data means **applying special techniques and tools to create meaningful visualizations from extremely large, fast and complex datasets** that normal visualization methods cannot handle.

Simply put — regular charts work for small data. Big Data needs special approaches to visualize effectively.

Why is Visualizing Big Data Different?

Regular Data	Big Data
Thousands of rows	Billions of rows
Static data	Real time streaming data
One data type	Multiple data types
Normal tools work	Special tools needed
Simple charts enough	Advanced techniques needed

5: Visualizing Big Data

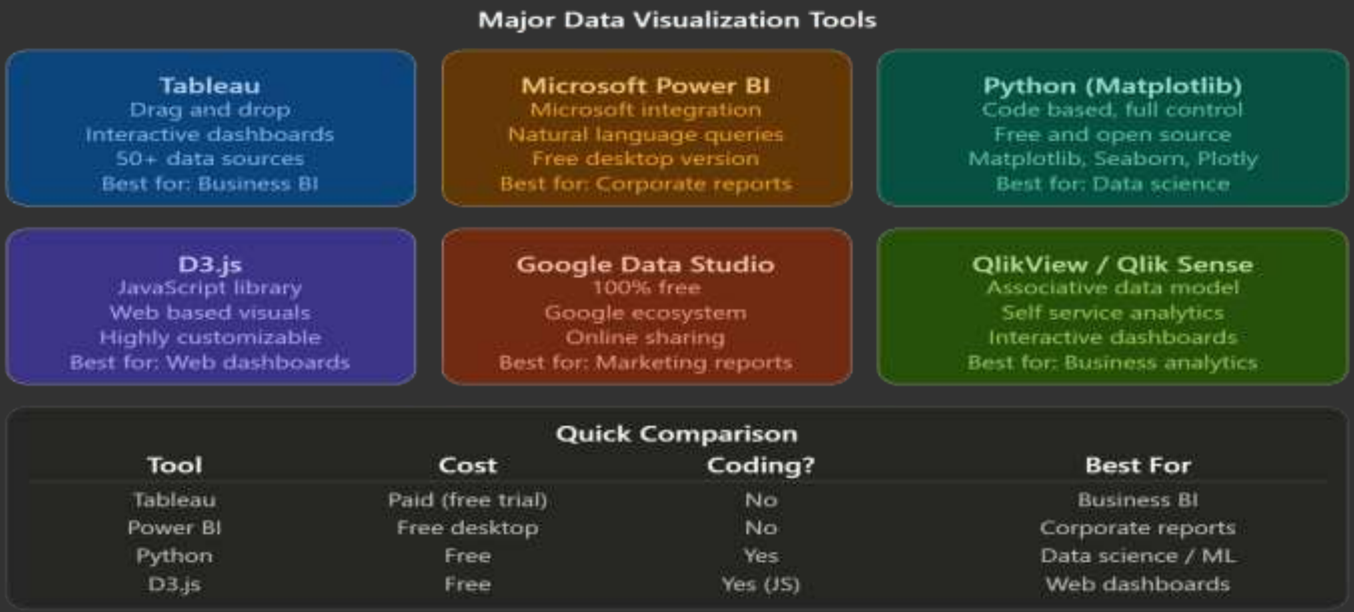


6: Tools Used in Data Visualization

What are Data Visualization Tools?

Data Visualization Tools are **software applications** that help users create charts, graphs, dashboards and other visual representations of data easily without writing much code.

Simply put — instead of drawing charts manually, these tools do it automatically from your data.



Major Data Visualization Tools

1. Tableau

What is Tableau? Tableau is a powerful business intelligence and data visualization tool that allows users to create interactive and shareable dashboards without any coding.

Key Features of Tableau:

Drag and Drop — no coding needed

Connects to 50+ sources — Excel, SQL, Google Sheets, cloud databases

Interactive dashboards — filters, drill downs

Real time data — live connection to databases

Mobile friendly — works on phones and tablets

Tableau Public — free version to share public visualizations

Collaboration — share dashboards with team online

Types of Tableau Products:

Tableau Desktop — create visualizations

Tableau Server — share within organization

Tableau Online — cloud based sharing

Tableau Public — free, public sharing

Tableau Reader — view Tableau files (free)

Applications of Tableau:

Sales performance dashboards

Customer behavior analysis

Financial reporting

Supply chain monitoring

Healthcare patient analytics

Social media analytics

2. Microsoft Power BI

Microsoft's data visualization tool — very popular in corporate world.

What it does:

Creates reports and dashboards

Integrates well with Microsoft products (Excel, Azure)

Free desktop version available

Share reports online via Power BI Service

Key Features:

Deep integration with Excel and Office 365

Natural language queries ("Show me sales by region")

Automatic insights from data

Free desktop version

Applications:

Corporate reporting

KPI monitoring

Sales and marketing analytics

3. Python (Matplotlib, Seaborn)

Programming based visualization using Python libraries.

What it does:

Create any type of chart using code

Full control over visualization

Free and open source

Key Libraries:

Matplotlib — basic charts (line, bar, scatter, histogram)

Seaborn — statistical visualizations built on Matplotlib

Plotly — interactive charts

Applications:

Data science projects

Research and academia

Machine learning model evaluation

4. D3.js

JavaScript library for creating interactive web visualizations.

What it does:

Creates highly customized interactive visualizations

Runs in web browser

Maximum flexibility

Applications:

Web based dashboards

Custom interactive charts

Data journalism

5. Google Data Studio

Free online tool by Google for creating reports and dashboards.

What it does:

Connects to Google Analytics, Google Sheets, BigQuery

Free to use

Share reports as web links

Applications:

Website analytics reporting

Marketing campaign tracking

SEO reporting

6. QlikView / Qlik Sense

Business intelligence tool similar to Tableau.

What it does:

Self service analytics

Associative data model — shows relationships between all data

Interactive dashboards

7: Hadoop Ecosystem

What is Hadoop?

Hadoop is an **open source framework** for storing and processing extremely large datasets (Big Data) across a cluster of computers.

Simply put — Hadoop allows you to store and process data that is too large for a single computer by **distributing the work across many computers** working together.

Key Features of Hadoop

Scalable — add more computers as data grows

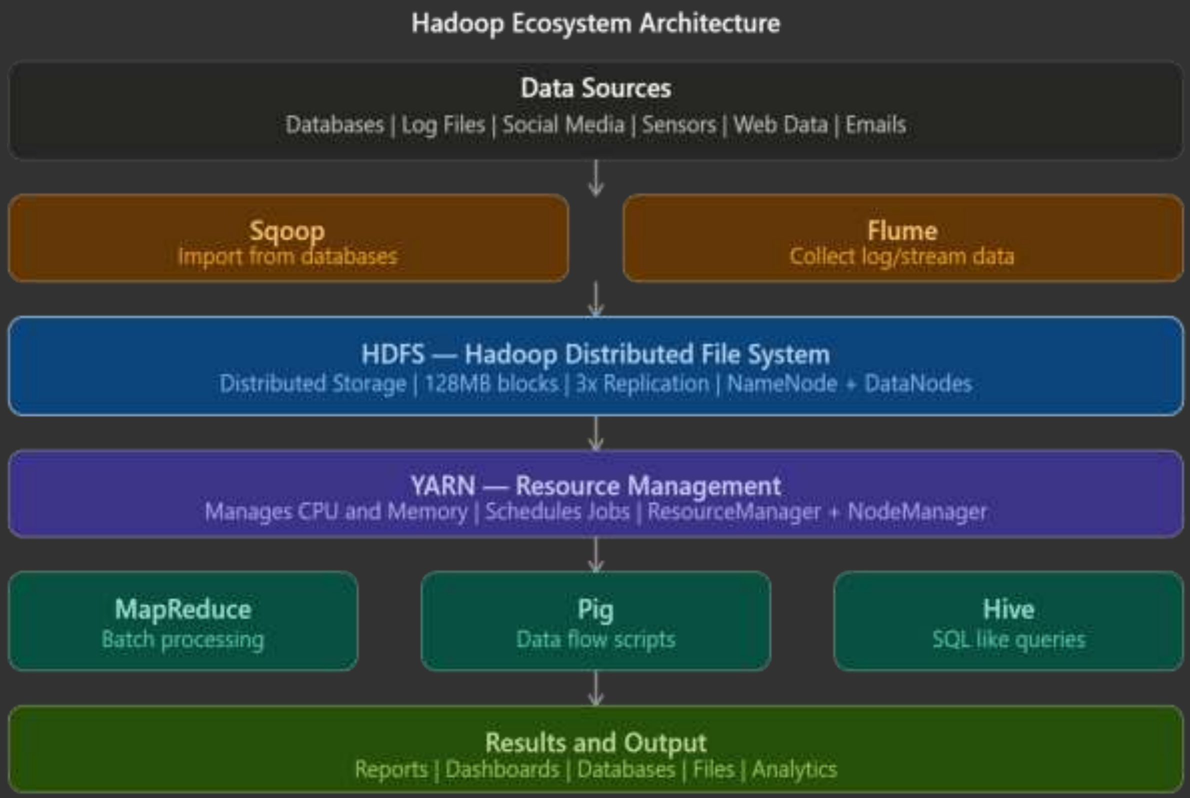
Fault Tolerant — if one computer fails, others continue

Cost Effective — runs on cheap commodity hardware

Flexible — handles structured, unstructured and semi-structured data

Open Source — free to use

7: Hadoop Ecosystem



Core Components of Hadoop

1. HDFS (Hadoop Distributed File System)

The **storage layer** of Hadoop.

Stores data across multiple computers

Data is split into **blocks** (default 128 MB each)

Has **NameNode** (master) and **DataNodes** (workers)

NameNode = manages file system metadata,

knows where data is stored

DataNode = actually stores the data blocks

2. MapReduce

The **processing layer** of Hadoop.

Processes data in parallel across many computers

Two phases:

Map (split and process) and **Reduce** (combine results)

3. YARN (Yet Another Resource Negotiator)

The **resource management layer** of Hadoop.

Manages and allocates resources (CPU, memory) across the cluster

Schedules jobs to run on available nodes

Acts as the operating system of Hadoop cluster

4. Hadoop Common

Utilities and libraries shared by all Hadoop modules.

Common Java libraries.

Supporting files needed by other Hadoop components.

8: MapReduce

What is MapReduce?

MapReduce is a **programming model** used to process and analyze large amounts of data in parallel across a cluster of computers.

Simply put — MapReduce breaks a big problem into smaller pieces, solves each piece separately (Map), then combines all results together (Reduce).

Two Phases of MapReduce

Phase 1: Map

Input data is **split into chunks**

Each chunk is processed by a **mapper** independently

Mapper produces **key-value pairs** as output

Multiple mappers run in **parallel.**

Phase 2: Reduce

All key-value pairs with **same key** are grouped together.

Reducer combines/aggregates the grouped values

Produces the **final output.**

8: MapReduce

Complete Example — Word Count

Input Text:

Doc1: "cat dog cat"

Doc2: "dog bird dog“

Map Phase: Each document is processed separately:

Doc1 mapper output:

(cat, 1), (dog, 1), (cat, 1)

Doc2 mapper output:

(dog, 1), (bird, 1), (dog, 1)

Shuffle and Sort: Groups same keys together:

(bird, [1])

(cat, [1, 1])

(dog, [1, 1, 1])

Reduce Phase: Adds up values for each key:

(bird, 1)

(cat, 2)

(dog, 3)

Final Output:

bird = 1

cat = 2

dog = 3

9: Pig

What is Apache Pig?

Apache Pig is a **high level platform** built on top of Hadoop that allows users to write simple scripts to process and analyze large datasets without writing complex MapReduce code.

Simply put — instead of writing complicated Java code for MapReduce, Pig lets you write simple scripts called **Pig Latin** to do the same job.

Key Features of Pig

Easy to learn — Pig Latin is simple scripting language

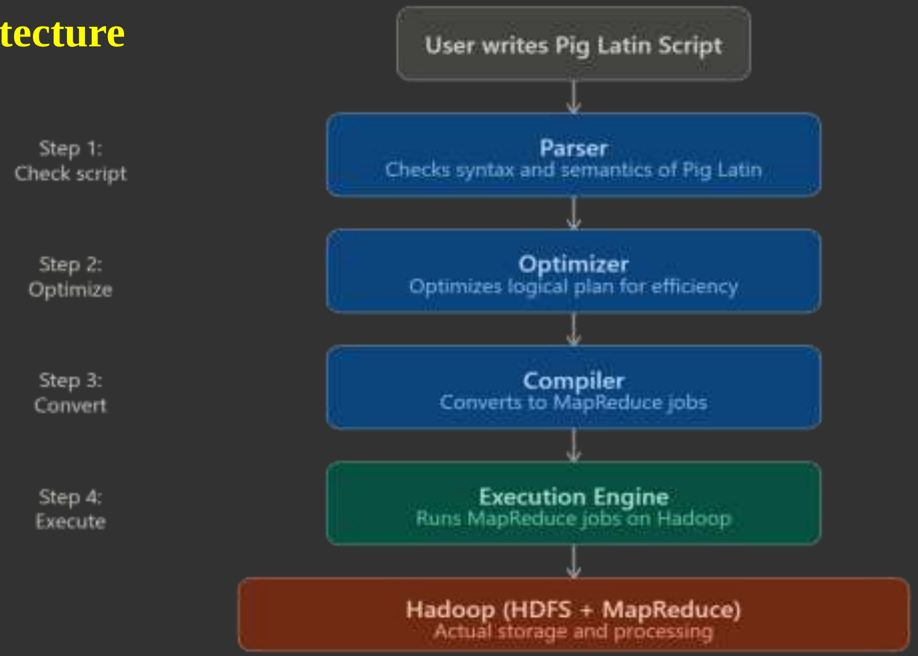
Less code — 10 lines of Pig Latin = 200 lines of Java MapReduce

Flexible — handles structured, semi-structured and unstructured data

Automatically optimized — Pig optimizes scripts automatically

Built on Hadoop — runs on top of HDFS and MapReduce

Pig Architecture



Real Life Applications of Pig

ETL (Extract Transform Load) — moving data between systems

Log processing — analyzing web server logs

Data cleaning — removing bad records from large datasets

Research — processing scientific data at Yahoo, Twitter

10: Hive

What is Apache Hive?

Apache Hive is a **data warehouse tool** built on top of Hadoop that allows users to **query and analyze large datasets using SQL-like language** called HiveQL.

Simply put — Hive lets you use SQL queries on Big Data stored in Hadoop HDFS. If you know SQL, you can query Big Data using Hive!

Key Features of Hive

SQL like language — HiveQL is very similar to SQL

Built on Hadoop — runs on top of HDFS and MapReduce

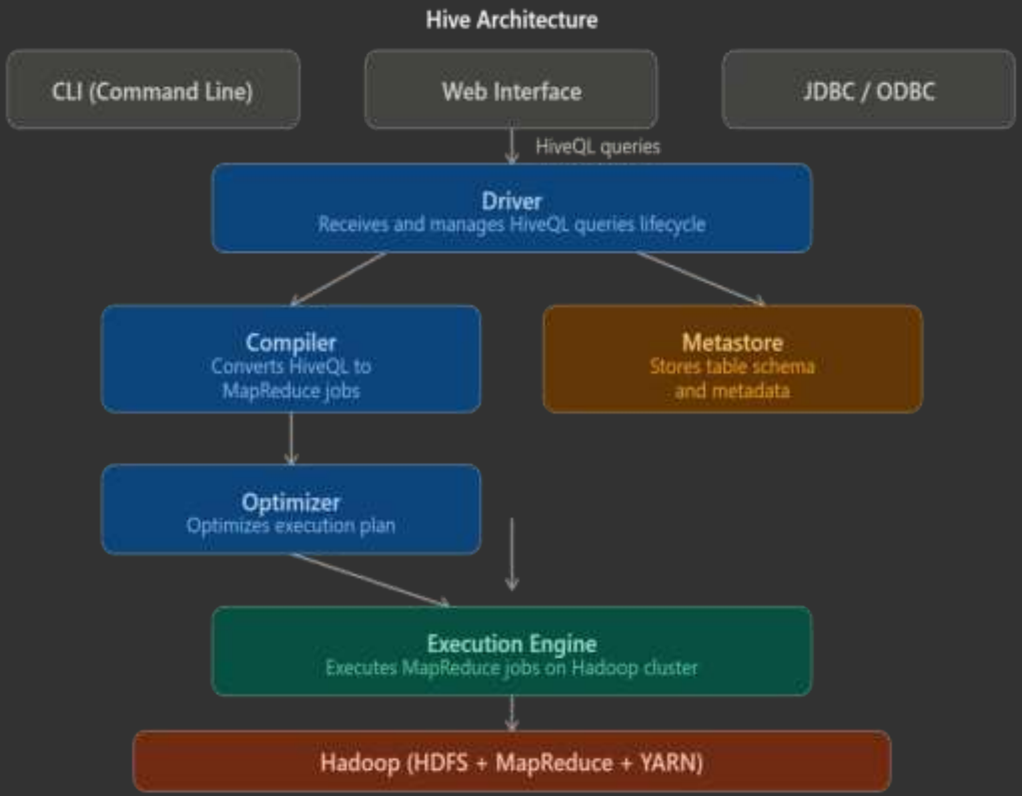
Schema on read — applies structure when reading data

Handles huge data — processes petabytes of data

Supports multiple formats — CSV, JSON, ORC, Parquet

Free and open source — developed by Facebook, now Apache

Hive Architecture



Hive Architecture Components Explained

1. User Interface (UI)

Ways to interact with Hive

CLI (command line), Web UI, JDBC/ODBC drivers

2. Driver

Receives HiveQL queries from user

Manages the complete lifecycle of query execution

3. Compiler

Converts HiveQL query into MapReduce jobs

Checks syntax and semantics

4. Metastore

Stores metadata — table names, column names, data types, locations

Acts like a database catalog

Usually uses MySQL database.

5. Optimizer

Optimizes the execution plan

Reduces unnecessary operations

6. Execution Engine

Actually runs the MapReduce jobs on Hadoop cluster

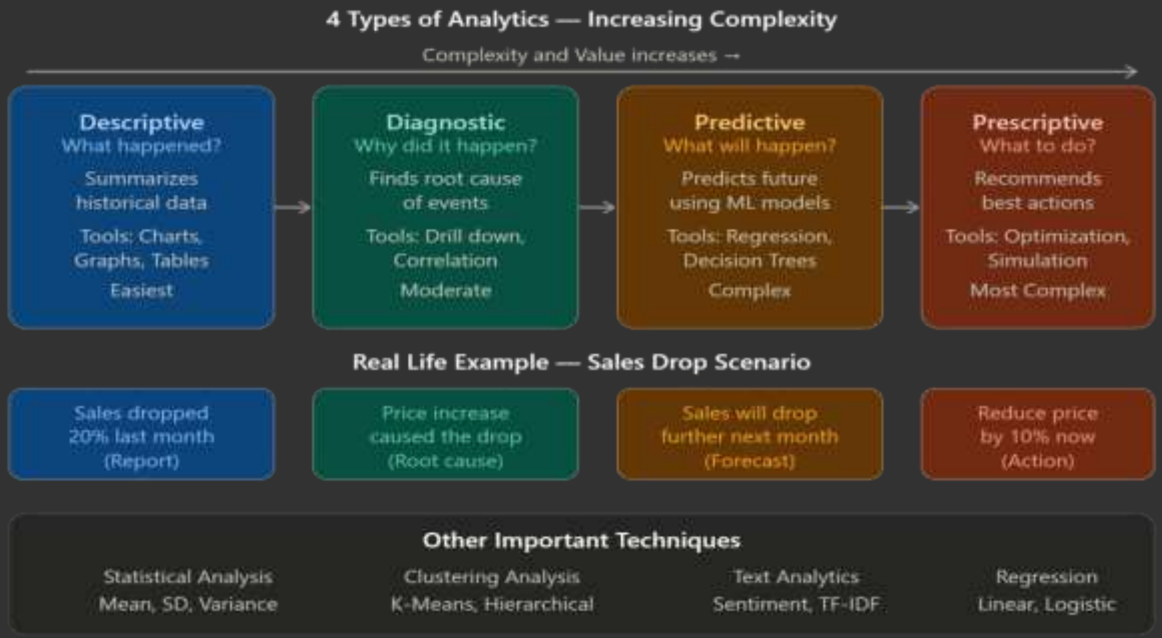
Returns results back to user

11: Analytical Techniques Used in Big Data Visualization

What are Analytical Techniques?

Analytical Techniques in Big Data Visualization are **methods used to analyze and extract meaningful insights from large datasets** and then present those insights visually.

Simply put — these are the "how to analyze" methods that help us understand Big Data before and after visualizing it.



12: Line Plot

What is a Line Plot?

A Line Plot is a type of chart that **displays data points connected by straight lines** to show trends and changes over time or across categories.

Simply put — dots connected by lines that show how values go up or down over time.

When to Use Line Plot

Data changes **over time.**

Showing **trends** — increasing, decreasing, fluctuating.

Comparing **multiple trends** on same graph.

Real Life Applications

Application	Example
Stock market	Stock price over 1 year
Weather	Temperature over a week
Sales	Monthly revenue comparison
Website	Daily visitors over a month
Health	Patient heart rate over time

13: Scatter Plot

What is a Scatter Plot?

A Scatter Plot is a type of chart that **displays individual data points as dots** on a 2D graph to show the **relationship (correlation) between two variables**.

Simply put — plot two variables on X and Y axis as dots and see if they are related or not.

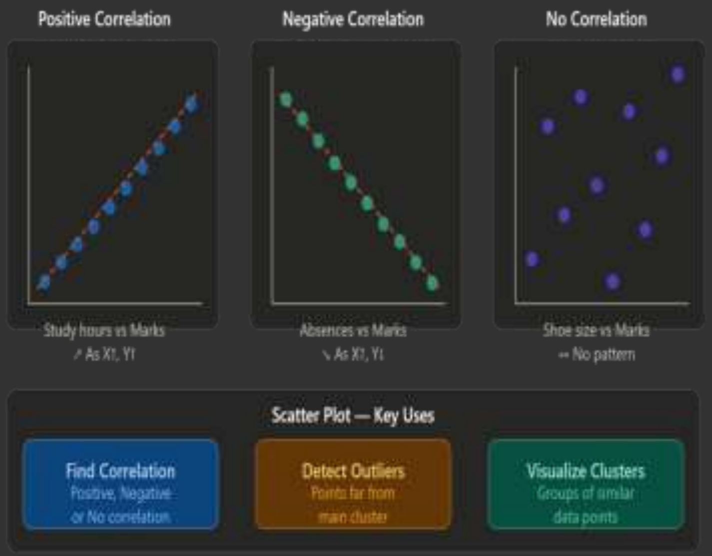
When to Use Scatter Plot

Finding **relationship** between two variables

Detecting **outliers** in data

Checking **correlation** before building ML models

Visualizing **clusters** in data



Real Life Applications

Application	Example
Education	Study hours vs exam marks
Healthcare	Age vs blood pressure
Business	Advertising spend vs sales
Sports	Practice hours vs performance
Real estate	House size vs price

14: Histogram

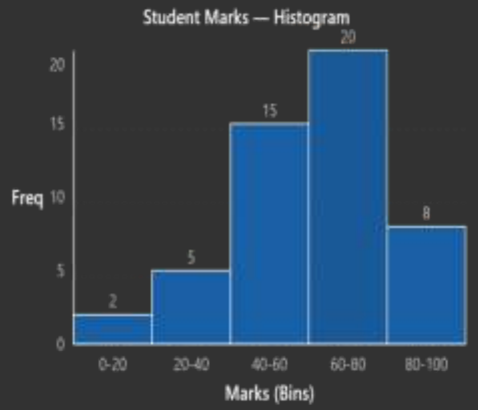
A Histogram is a type of bar chart that **shows the distribution (frequency) of a single continuous variable** by dividing data into intervals called **bins**.
Simply put — histogram shows how many data points fall in each range. It answers: "How is my data distributed?"

When to Use Histogram

To see **distribution** of data
To find **skewness** (is data spread evenly or tilted?)
To detect **outliers**
To understand **central tendency** (where most data lies)
For single **continuous** variable

Real Life Applications

Application	Example
Education	Distribution of student marks
Healthcare	Distribution of patient ages
Business	Distribution of product prices
Finance	Distribution of daily stock returns
Manufacturing	Distribution of product weights



15: Density Plot

A Density Plot is a **smoothed version of a histogram** that shows the **distribution of a continuous variable** as a smooth curve instead of bars.

Simply put — instead of showing frequency as bars like histogram, density plot shows it as a smooth flowing curve.

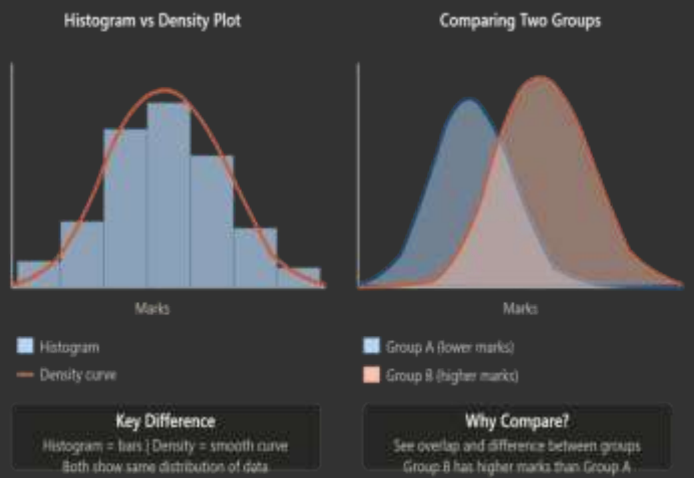
When to Use Density Plot

To see **distribution** of continuous data (like histogram)

When you want a **smoother** visualization than histogram

To **compare distributions** of multiple groups

When exact bin size should not affect the visualization



Real Life Applications

Application	Example
Education	Compare marks distribution of two classes
Healthcare	Compare age distribution of two patient groups
Finance	Compare returns distribution of two stocks
Manufacturing	Compare weight distribution before and after process change

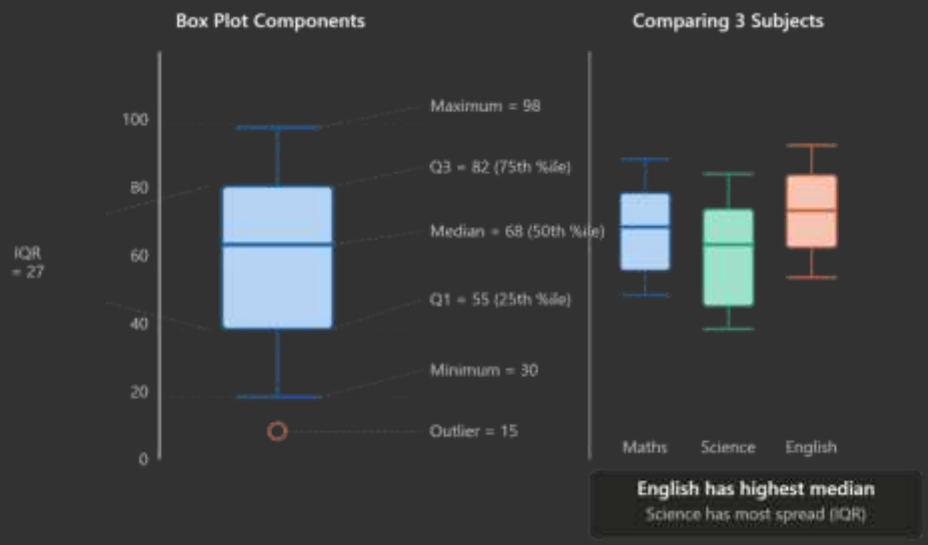
16: Box Plot

A Box Plot (also called **Box and Whisker Plot**) is a chart that shows the **statistical summary of data** using 5 key values — Minimum, Q1, Median, Q3 and Maximum.

Simply put — a box plot gives you a complete picture of how data is spread in just one diagram.

5 Key Components of Box Plot

Component	Meaning
Minimum	Smallest value (excluding outliers)
Q1	25th percentile — 25% data below this
Median	Middle value — 50% data below this
Q3	75th percentile — 75% data below this
Maximum	Largest value (excluding outliers)



Real Life Applications

Application	Example
Education	Compare marks across subjects
Healthcare	Compare patient age across hospitals
Business	Compare salary across departments
Finance	Compare stock returns across sectors
Manufacturing	Compare product quality across factories

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